

Title: Design of a Fault-Tolerant ARINC 429 Communication Controller ASIC

Supervisor

Mr. S. Senthil Rajan

Assistant Professor, Department of ECE

Batch Number : B1

Members of students :

1. Kishore M - (Reg. No.: 2127230701065)
2. Mukesh K - (Reg. No.: 2127230701083)
3. Pranay V - (Reg. No.: 2127230701104)

Presentation Outline

- Project Title and Domain
- Problem Statement
- Objectives of the Project
- Proposed Solution and Methodology
- Innovation and Novelty
- Feasibility (Technical, Financial, Time-bound)
- Tools, Technologies, and Components Planned
- Expected Outcomes
- Applications and Future Scope
- Work Distribution among Team Members
- Phase-wise Execution Plan



Project Title:

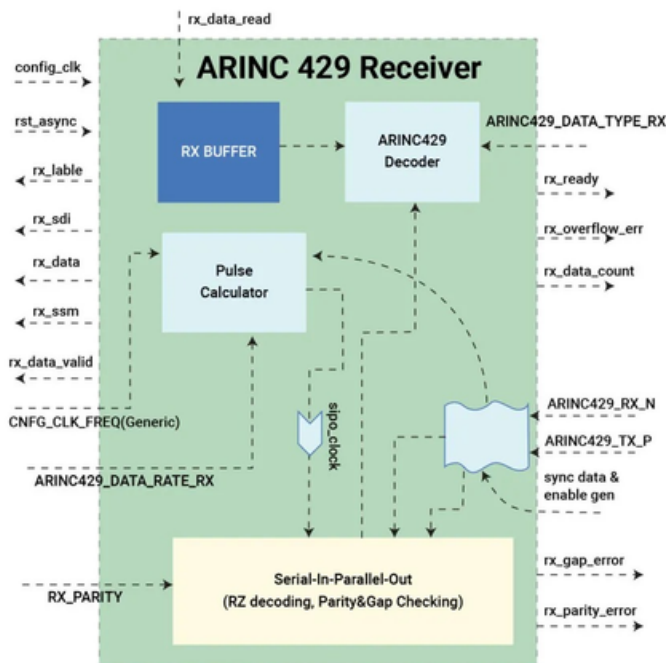
Design of a Fault-Tolerant ARINC 429 Communication Controller ASIC

Domain: VLSI



Problem Statement

Conventional ARINC 429 communication controllers are **designed to transmit and receive data with basic error detection, such as parity checking**. However, they cannot continuously monitor the health of the communication channel or distinguish between temporary and repeated communication faults. This can reduce the reliability of aircraft communication systems. Therefore, **there is a need for a hardware-based communication controller** that can detect faults, monitor communication health, and automatically perform suitable recovery actions to improve reliability and system performance.



Key Limitations:

- Limited Error Detection
- No Health Monitoring
- No Fault Classification
- No Adaptive Recovery
- Reduced Communication Reliability

Objectives of the Project

The project focuses on identifying and monitoring critical communication faults that can affect the reliability of the ARINC 429 communication channel. The **following parameters are continuously monitored** to assess communication health and enable intelligent fault detection and adaptive recovery.

Parity Error

Occurs when the received parity bit does not match the transmitted data, indicating data corruption.



Communication Timeout

No valid data is received within the expected time limit.



FIFO Overflow

Incoming data exceeds the FIFO storage capacity, causing data loss.



FIFO Underflow

The FIFO becomes empty while data transmission is still in progress.



Invalid Label Error

A received ARINC 429 label is unknown or not supported by the controller.



Repeated Communication Error

The same error occurs continuously, indicating a persistent communication fault.



Bus Idle Error

No communication activity is detected on the ARINC 429 bus for a prolonged period.

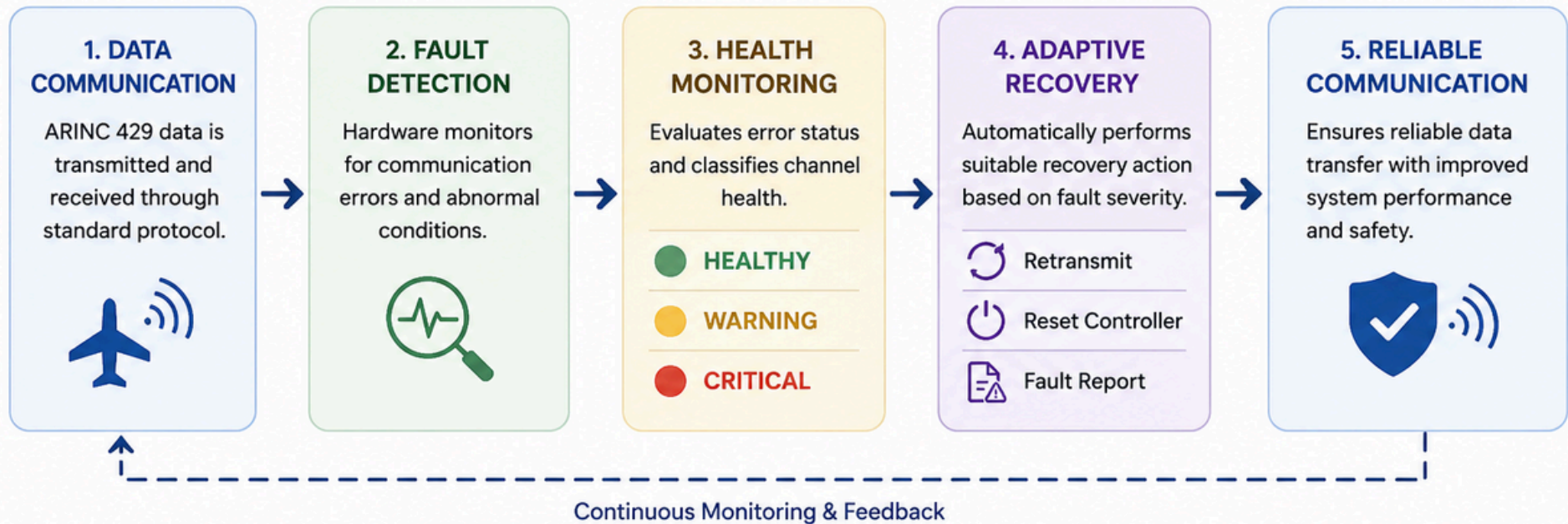


High Error Rate

The number of communication errors exceeds a predefined threshold, indicating channel degradation.



Proposed Solution/Methodology

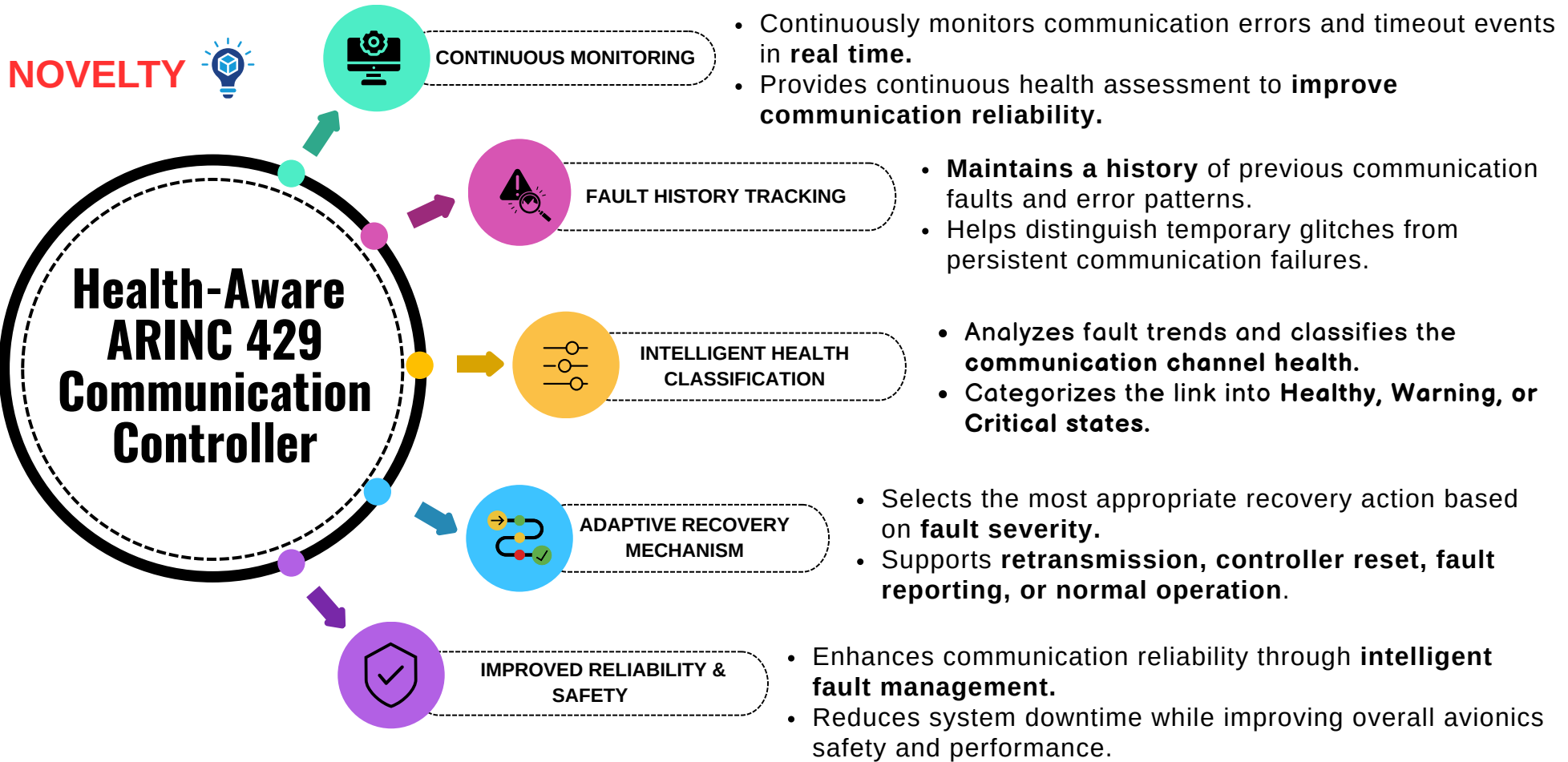


Project Implementation Methodology:

- **Protocol Analysis** – Analyze ARINC 429 protocol.
- **RTL Architecture Design** – Develop controller architecture.
- **Verilog HDL Coding** – Implement digital hardware.
- **Simulation & Verification** – Perform functional verification.
- **FPGA Implementation** – Validate hardware prototype.
- **ML Model Development** – Train fault prediction model.
- **Performance Evaluation** – Evaluate area, power, timing, and prediction accuracy.



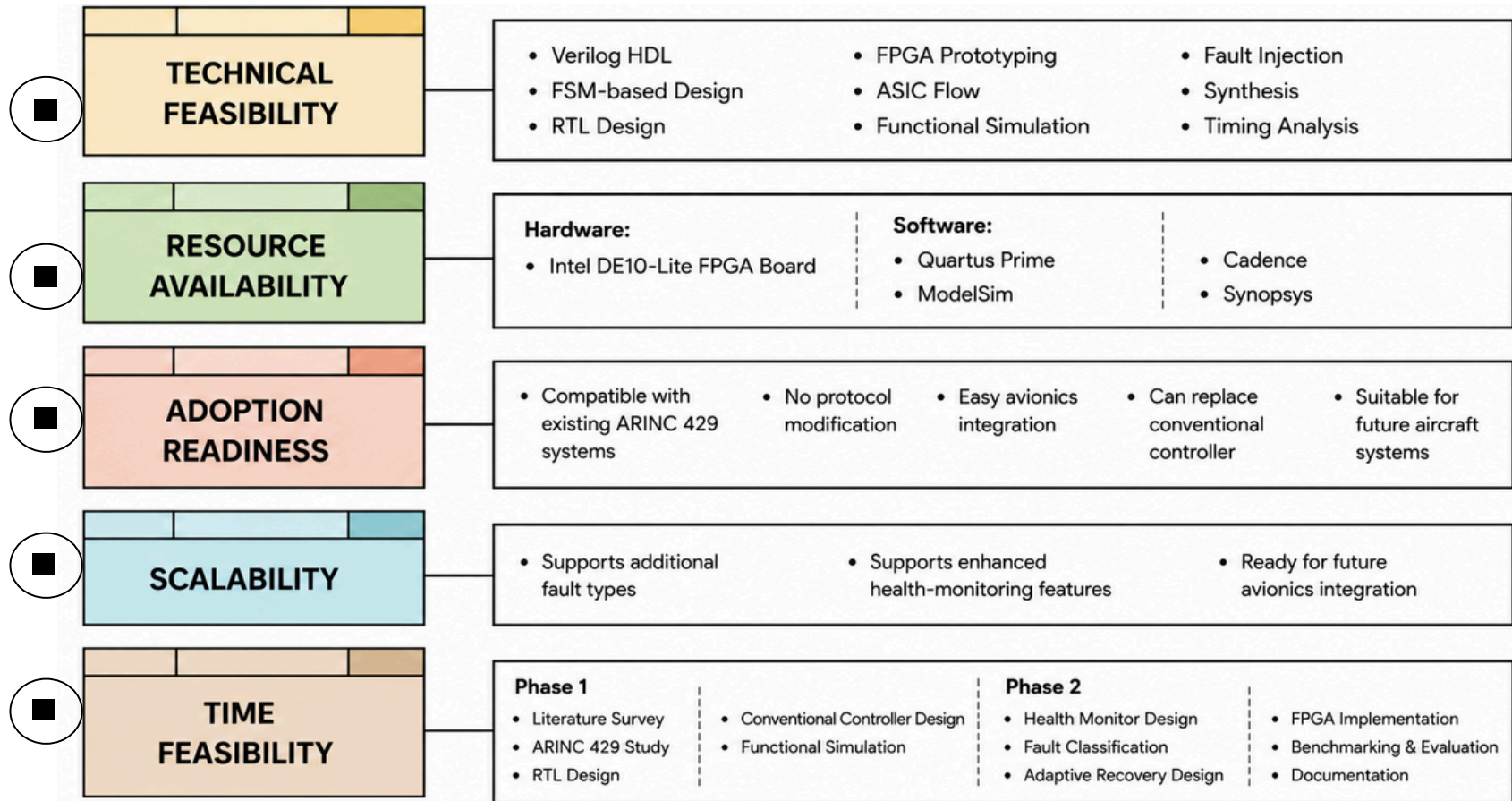
Novelty & Innovation



INNOVATION

Introduces a hardware-based health-aware architecture that **continuously monitors communication status, analyzes fault history, and dynamically selects** the optimal recovery action to **improve system reliability**.

Feasibility



The proposed Health-Aware ARINC 429 Communication Controller is **technically feasible** as it uses the standard **ARINC 429 protocol**, **Verilog HDL**, **FPGA platforms**, and **commonly available VLSI design tools**. The modular design ensures simple implementation with minimal hardware overhead, making the solution practical and cost-effective.



Tools, Technologies, and Components Planned

★ Hardware Components:

- **Intel DE10-Lite FPGA Board:** RTL implementation, FPGA prototyping, and hardware validation.
- **Computer / Workstation:** Design, simulation, synthesis, timing analysis, and documentation.

★ Communication & Hardware Modules:

- **ARINC 429 Transmitter & Receiver:** Data transmission, reception, and decoding.
- **Parity Generator & Checker:** Error detection using odd parity.
- **Timeout Detection Unit:** Detects communication delays and missing messages.
- **Health Monitoring Unit:** Tracks communication status and fault events.
- **Fault Classification Engine:** Identifies transient and persistent faults.
- **Adaptive Recovery Controller:** Performs retransmission, reset, or fault reporting.

★ Software & Design Technologies:

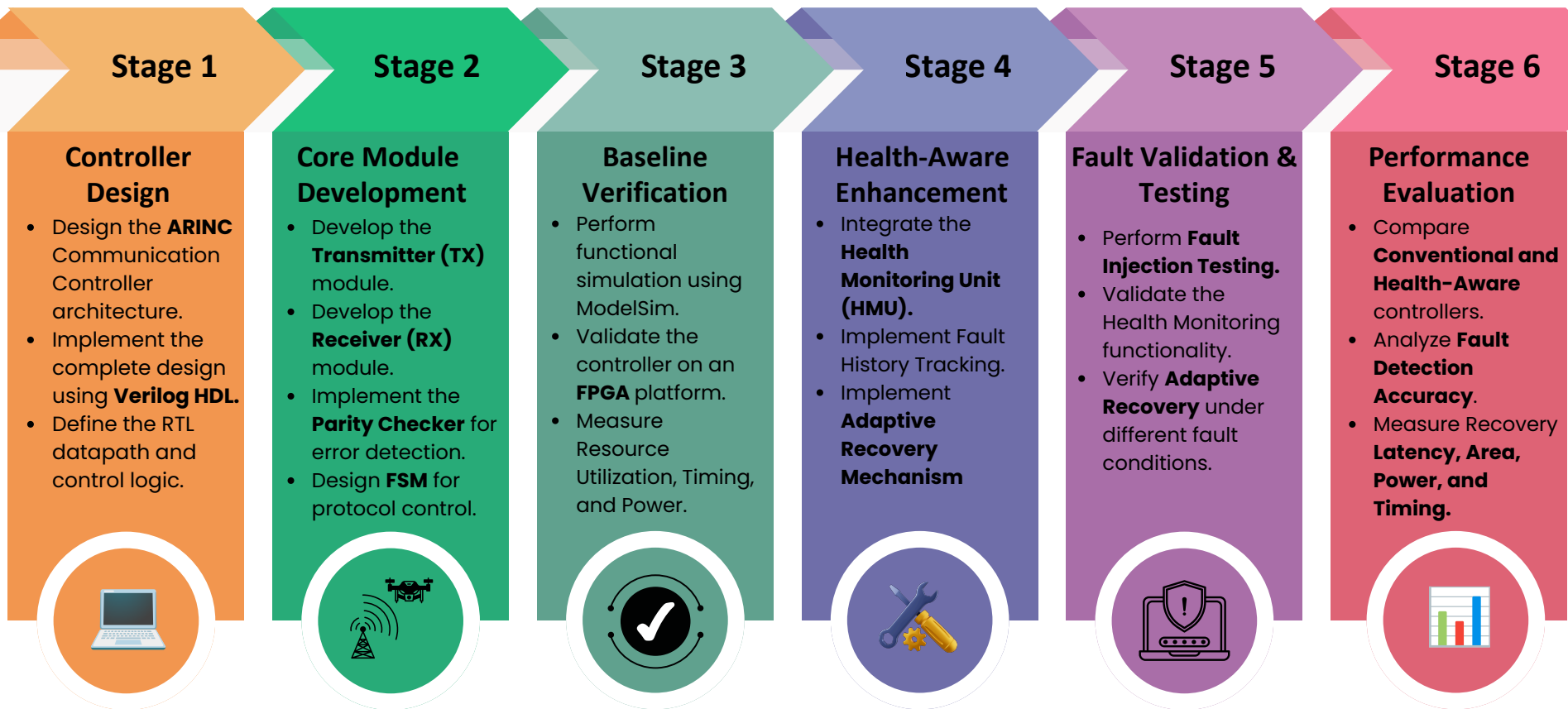
- **ARINC 429:** Standard avionics serial communication protocol.
- **Verilog HDL:** Hardware description language for RTL implementation.
- **RTL & FSM-Based Design:** Digital design methodology using Register Transfer Level and Finite State Machines.
- **FPGA Prototyping:** Hardware implementation and functional validation on Intel DE10-Lite FPGA.
- **RTL-to-GDSII ASIC Design Flow:** Complete ASIC implementation flow from RTL design to physical layout.

★ Development & Verification Tools:

- **Intel Quartus Prime:** FPGA synthesis, implementation, resource utilization, and timing analysis.
- **ModelSim:** Functional simulation and RTL verification.
- **Cadence:** ASIC physical design and layout.
- **Synopsys:** Logic synthesis and static timing analysis.



Expected Outcomes



Performance Optimization Objectives:

- **Minimized logic** utilization for efficient hardware implementation.
- **Reduced power consumption** through optimized RTL design.
- **Lower critical path delay** for improved operating speed.
- **Fast fault detection** with adaptive recovery support.



Phase-wise Execution Plan

PHASE 1

Communication Controller & ML Development

- Literature Survey & ARINC 429 Protocol Analysis
- System Architecture Design
- RTL Design using Verilog HDL
- ARINC 429 Transmitter & Receiver Development
- Functional Simulation & Verification
- ML Dataset Preparation & Model Development
- FPGA Implementation & Baseline Validation

PHASE 2

Health Monitoring & System Integration

- Health Monitoring Unit Design
- Fault Detection Unit Development
- Fault History Register Implementation
- ML Model Integration with Health Monitor
- Adaptive Fault Recovery Controller Design
- Fault Injection & Functional Verification
- FPGA Validation & Performance Evaluation
- Benchmark Comparison with Conventional Controller

Project Duration: Approximately 7-8 Months

Applications:

- **Commercial Aircraft** – Reliable communication
- **Military Aviation** – Mission-critical systems
- **Avionics Systems** – Data exchange
- **Aircraft Monitoring** – Health monitoring
- **FPGA Prototyping** – Hardware validation
- **ASIC Development** – Chip implementation
- **Aerospace Industry** – Industrial deployment

Future Scope:

- **AI Integration** – Intelligent prediction
- **Multi-Channel Support** – Parallel communication
- **Protocol Extension** – Advanced compatibility
- **ASIC Fabrication** – Commercial deployment
- **Hardware Security** – Secure communication
- **Low-Power Design** – Energy efficiency
- **Real-Time Analytics** – Smart monitoring

Research & Publication:

- We plan to publish this work in a reputed journal and explore opportunities for patent filing based on the project's novel contributions.



References

S. No.	Paper Title	Link
1	Development of SoC for Interfacing Avionics Units with ARINC 429 Bus	https://www.researchgate.net/publication/306392847_Development_of_SoC_for_interfacing_avionics_units_with_ARINC_429_bus
2	Implementation and Performance Analysis of Multi-Channel ARINC 429 Controller on FPGA	https://acikerisim.gtu.edu.tr/items/b670c883-cd73-4992-8271-6df0696b4c4d
3	VHDL Implementation and Verification of ARINC-429 Core	https://arxiv.org/abs/1011.5374
4	ARINC-429 Airborne Communications Transceiver System Based on FPGA Implementation	https://www.researchgate.net/publication/300616811_ARINC_-_429_airborne_communications_transceiver_system_based_on_FPGA_implementation
5	Hardware Fingerprinting for the ARINC 429 Avionic Bus	https://arxiv.org/abs/2003.12456
6	ARINC 429 Cyber-Vulnerabilities and Voltage Data in a Hardware-in-the-Loop Simulator	https://arxiv.org/abs/2408.16714

Analyzed 9 Journal Papers & Research Articles.

Team Work Distribution

Pranay V



ARINC 429 & RTL Design

Roles & Responsibilities:

- Literature Survey
- ARINC 429 Protocol Analysis
- RTL Architecture Design
- Verilog HDL Implementation of Transmitter & Receiver
- Communication Controller Integration
- RTL Functional Validation
- Module Integration
- Documentation

Kishore M



Fault Detection & FPGA Implementation

Roles & Responsibilities:

- Literature Survey
- Fault Detection Unit, Health Monitoring & Adaptive Recovery FSM
- Functional simulation and fault injection
- FPGA implementation and validation
- Area, Power & Timing analysis
- Documentation

Mukesh K



ML Integration & Testing

Roles & Responsibilities:

- Literature Survey
- ML-based communication health prediction
- Dataset preparation and model evaluation
- Hardware logic mapping and testing
- Presentation preparation and final demonstration
- Documentation

All team members will work collaboratively, coordinate effectively, and contribute throughout the project lifecycle.

- Clarity and relevance of problem statement
- Novelty and originality
- Technical feasibility
- Structured planning and team coordination
- Realistic and achievable goals in both phases

